CareBridge

a micro project e-report Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the micro project e-report titled **"CareBridge"** is the bonafide work of "Demian Sharwin Xaxa & Nithya Maria Mathew**"** who carried out the project work under my supervision.

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ABSTRACT

People who leave hospital often go home with a discharge summary or prescription written in dense clinical English, full of abbreviations such as OD, BD, PC and SOS. Mistakes in taking the listed medicines after discharge are a recognised source of avoidable harm, and the problem is worse for patients and family carers who read Tamil or Hindi more comfortably than English. **CareBridge** is a prototype web application that tries to close this gap. The user photographs a discharge summary or prescription; a locally run vision-language model (Qwen3-VL 8B through Ollama) reads the medicine list; a rule layer normalises the result and decides, row by row, whether the user must confirm it; the app then explains each medicine in plain English, Tamil or Hindi, checks medicine pairs against the DDInter interaction database, and builds a daily dose schedule and a one-page summary for the doctor.

The design follows one principle: when the system is unsure it must say so, and it must never claim that a combination is safe. Pairs it cannot check are shown as “not checked”. Everything runs on the user’s own computer, and a public GitHub Pages site shows only a pre-recorded sample.

The system was evaluated on synthetic documents so that the ground truth is known. A held-out test split of 40 documents, each rendered in five image conditions (795 medicine mentions), was run once under a frozen configuration. After the rule layer, medicine-level F1 was 0.999 (raw model output 0.984) and 88.8% of medicine entries were fully correct (raw 26.2%). No wrong generic name reached the interaction check (0 of 389). Two of 285 schedule-ready medicines had a wrong schedule, and the review layer asked the user to confirm 51.1% of medicines. English explanation lines were verified 93.7% of the time, but Tamil only 18.0% (76.1% fell back to English), so Tamil support is weak. A separate stress set of 30 authored prescriptions met all four pre-registered criteria. The report states the limits plainly: synthetic data, no clinical validation, no pharmacist or native-speaker review, and the same model both writes and judges explanations.

Keywords: post-discharge care, medication safety, vision-language model, multilingual health communication, drug–drug interaction, local-first AI, evaluation under a frozen protocol.

TABLE OF CONTENTS

S.No.	Title		Page No.
	BONAFIDE CERTIFICATE		ii
	ABSTRACT		iii
	TABLE OF CONTENTS		iv
1	**INTRODUCTION**		1
	1.1	Background of the selected problem	1
	1.2	Industry Relevance	1
	1.3	Aim and Objective of the Micro Project	2
	1.4	Link to SDG / KITS Technology Mission	2
2	**LITERATURE REVIEW**		4
	2.1	Medication safety after discharge	4
	2.2	Health literacy and language access	4
	2.3	Vision-language models for document reading	4
	2.4	Hallucination, verification and LLM judges	4
	2.5	Drug–drug interaction knowledge bases	5
	2.6	Evaluation methodology	5
	2.7	Gap addressed by this project	5
3	**METHODOLOGY**		6
	3.1	Development approach	6
	3.2	Design principles	6
	3.3	System architecture and technology stack	6
	3.4	Processing pipeline	7
	3.5	Data: synthetic documents and splits	8
	3.6	Interaction-safety method	9
	3.7	Evaluation design and metrics	10
	3.8	Privacy and security design	11
	3.9	Hosting and deployment	11
4	**FABRICATION/SIMULATION/PROGRAM OR CODE/DESIGN**		12
	4.1	Repository and code organisation	12
	4.2	Extraction service	12
	4.3	Normalisation rules and review levels	12
	4.4	Confirmation overlay and formulary picker	13
	4.5	Explanation service with verification	13
	4.6	Safety engine and scheduler	14
	4.7	User interface design	14
	4.8	Software testing and quality checks	14
5	**RESULTS AND INTERPRETATION**		16
	5.1	Experimental setup	16

S.No.	Title		Page No.
	5.2	Medicine extraction	16
	5.3	Review burden and readiness	17
	5.4	Interaction safety	17
	5.5	Explanation in three languages	18
	5.6	Post-hoc case review	18
	5.7	Development versus test comparison	19
	5.8	Cluster-aware intervals	20
	5.9	Stress test on authored prescriptions	20
	5.10	Interpretation	21
6	**CONCLUSION AND FUTURE SCOPE**		23
	6.1	Conclusion	23
	6.2	Limitations	23
	6.3	Future scope	23
	REFERENCES		25
	APPENDIX: Abbreviations and project links		26

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SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY
EVALUATION REPORT

S.No	Rubrics	Maximum Marks	Marks Obtained
1	Online Certification Completion	10	
2	Evaluation of Problem statement with SDG	5	
3	Methodology Implementation	10	
4	Result Analysis	5	
5	Report	10	
	Total	40	

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1 INTRODUCTION

1.1 Background of the selected problem

A hospital stay ends with a piece of paper: a discharge summary or a prescription. For the patient it is the only instruction for the weeks that follow. It lists drugs by brand or generic name, uses shorthand such as OD (once daily), BD (twice daily), PC (after food) and SOS (only if needed), and is almost always written in English. A patient who reads Tamil or Hindi more comfortably, an elderly parent managed by a son or daughter, or anyone recovering and tired, can easily misread a dose, mix up two similar-sounding names or take two drugs that should not be combined.

This is not a minor inconvenience. A study of adults discharged from a hospital medical service found that adverse events after discharge were common and that many were linked to medicines [1]. The World Health Organization made medication safety the theme of its third Global Patient Safety Challenge, “Medication Without Harm”, which aims to cut severe, avoidable medication-related harm [2]. Low health literacy is associated with poorer health outcomes [3], and instructions the patient cannot read comfortably are one route to that gap.

The problem selected for this micro project is therefore narrow and practical: **help a person understand and follow the medicine list on their own discharge paper, in a language they choose, and tell them honestly what could not be checked.** The project does not try to diagnose, to prescribe, or to replace a doctor or pharmacist.

1.2 Industry Relevance

Digital health companies, hospital IT teams and health-insurance providers are all interested in post-discharge tools, because poor adherence causes readmissions and avoidable cost. Three industry trends make this project relevant:

- **Document understanding with vision-language models.** A single model can now read a photographed page, including typed and handwritten-style text, without a separate OCR engine. This removes a large engineering step from health-document products.
- **Local-first and privacy-preserving AI.** Discharge papers contain personal health data. Running the model on the user’s own machine, as CareBridge does through Ollama, avoids sending that data to a cloud service. This matters for regulators, hospitals and patients alike.
- **Safety engineering around AI output.** Industry practice for clinical AI is moving from “a model answered” to “a model answered, a rule layer checked it, and the product showed its uncertainty”. CareBridge is built around this layered approach and reports how often each layer helped.

The multilingual dimension is specific to India. English-only medicine information excludes many users, and a product that supports Tamil and Hindi, with honest labelling of machine-drafted text, addresses a real market gap.

1.3 Aim and Objective of the Micro Project

Aim. To design, build and honestly evaluate a multilingual (English, Tamil, Hindi) post-discharge companion that reads a photographed medicine list, explains it in plain language, checks medicine pairs for known interactions and produces a daily schedule, while never claiming more certainty than it has.

Objectives.

1. Extract the medicine list from a photographed discharge summary or prescription using a local vision-language model.
2. Add a rule layer that corrects predictable model errors and marks every medicine as needing confirmation, a note, or nothing.
3. Let the user confirm or correct uncertain medicines through a formulary picker before any interaction check or schedule is produced.
4. Explain each medicine in English, Tamil or Hindi, verify each explanation line, and fall back to the original English when verification fails.
5. Check medicine pairs against a published interaction database and report “not checked” instead of “safe” whenever a pair cannot be checked.
6. Evaluate the whole chain under a frozen, pre-registered protocol on held-out synthetic documents, and report failures as clearly as successes.
7. Publish a verified, pre-recorded sample on a public static site so the work can be seen without any server.

Scope and non-goals. CareBridge is a class-project prototype. It has not been clinically validated, it is not a medical device, and it was tested on synthetic documents. Tamil and Hindi interface text is machine-drafted and has not been reviewed by a native speaker; the small author-checked interaction supplement has not been reviewed by a pharmacist. These limits are stated inside the application as well as in this report.

1.4 Link to SDG / KITS Technology Mission

SDG 3 – Good Health and Well-being. The project supports the aim of reducing avoidable harm and improving access to understandable health information (SDG 3, in particular the targets on quality essential health-care services and on strengthening early warning and risk management) [12]. Better understanding of discharge medicines is a low-cost contribution to safer recovery at home.

SDG 10 – Reduced Inequalities. Language is a barrier to care. Offering the same explanation in Tamil and Hindi, and being open about where translation quality is weak, is a small step toward equal access to medical information.

SDG 9 – Industry, Innovation and Infrastructure. The project shows that a useful AI health tool can run on a single consumer laptop with open models and open data, without a cloud service or a subscription.

KITS technology mission. The work falls under the institute's healthcare-and-technology focus: it applies artificial intelligence to a health problem and keeps the solution affordable, local and privacy-preserving.

2 LITERATURE REVIEW

This chapter reviews the work that CareBridge builds on. It is organised by the question each strand of work answers, and ends with the gap that this project addresses.

2.1 Medication safety after discharge

Forster and colleagues followed patients after discharge from a general medicine service and found that a substantial share suffered an adverse event, with drug-related events among the commonest [1]. The finding that harm continues after the patient leaves the ward is the starting point of this project. The WHO “Medication Without Harm” challenge generalises the concern and names medication errors at transitions of care as a priority area [2]. These sources justify the need but do not provide a tool.

2.2 Health literacy and language access

Berkman et al. reviewed the evidence on health literacy and concluded that lower literacy is associated with worse outcomes, including more hospitalisation and poorer use of medicines [3]. Written instructions that use abbreviations and a single language are therefore a risk factor in their own right. Machine translation and large language models can produce readable text in many languages, but the quality for lower-resource languages is uneven. CareBridge treats this as a measured quantity: it reports how many Tamil and Hindi lines survive verification instead of assuming good translation.

2.3 Vision-language models for document reading

Vision-language models (VLMs) accept an image and a text prompt and return text. The Qwen3-VL family is an open-weight series with strong document understanding, and its 8-billion-parameter instruct variant can be served locally [5], here through Ollama [11]. Earlier pipelines used OCR followed by rule-based parsing, which is brittle for photographs. A VLM can read the layout directly, but it may silently “correct” a misspelt drug name or fill an empty field from another part of the page. The evaluation in Chapter 5 measures exactly these behaviours.

2.4 Hallucination, verification and LLM judges

Surveys of hallucination in generated text describe how fluent output can contain content unsupported by the source [6]. Large language models can encode a good deal of clinical knowledge [10] but that does not make their free-text answers reliable enough to act on. A common mitigation is to have a second model pass judge the first. Zheng et al. showed that LLM judges agree well with humans on average but also exhibit biases, including favouring text written by the same model [7]. CareBridge uses verification on explanation lines but states the limitation openly: the same local model writes and judges, and no native speaker has reviewed the text.

2.5 Drug–drug interaction knowledge bases

DDInter is a curated, openly downloadable database of drug–drug interactions with severity levels and management notes [4]. It is a suitable source for a student project because it is public and structured. Its coverage is incomplete: in this project, more than half of the pairs among the 66 drugs in the app's formulary have no DDInter record. A responsible interface must therefore distinguish "an interaction was found", "no record exists" and "a medicine was not confirmed". Chapter 3 describes how CareBridge does this.

2.6 Evaluation methodology

Small test sets make point estimates unreliable. The Wilson score interval gives a better confidence interval for a proportion than the simple normal approximation [8], and the bootstrap of Efron resamples the data to estimate uncertainty without distributional assumptions [9]. Because several medicines come from the same document, they are not independent, so a bootstrap that resamples whole documents (a cluster bootstrap) is more honest than Wilson intervals on pooled mentions. CareBridge reports both. Good practice also separates tuning data from test data and fixes the protocol before the test is run; the project follows this with a frozen manifest and a one-shot official run.

2.7 Gap addressed by this project

Table 2.1: Summary of related strands and the gap addressed

Strand	What it provides	What is missing for a patient companion
Post-discharge safety studies [1][2]	Evidence that harm is real and common	No tool for the patient at home
Health-literacy research [3]	Link between comprehension and outcomes	No multilingual, document-based explanation
Vision-language models [5]	Reads photographed pages directly	Silent name corrections; no uncertainty signalling
LLM verification and judging [6][7]	Methods to check generated text	Same-model bias; weak in Tamil
DDInter [4]	Structured interaction data with severity	Incomplete coverage; no "not checked" semantics
Evaluation statistics [8] [9]	Intervals for small samples	Often not applied to clustered medicine data
CareBridge (this work)	End-to-end, local, multilingual chain with row-level review, "not checked" status and a frozen evaluation	Prototype only; synthetic data; unreviewed Tamil/Hindi

CareBridge's contribution is not a new model. It is the integration of these ideas into one working chain, with the safety behaviour designed first and the failures measured and reported openly.

3 METHODOLOGY

3.1 Development approach

The project was built in numbered phases, from planning and a licence check, through model selection and extraction (Phase 2), the normalisation, safety and explanation layers, the user interface (Phase 6), a frozen final evaluation (Phase 7) and public release (Phase 8). Each phase ended with a written report and a full automated test run before the next began. An AI coding assistant (Claude Code) was used under the author's direction to write and refactor code; every phase was gated by the tests and by a written report that listed defects found, deviations and open risks. Decisions are recorded in the repository's DECISIONS and PROGRESS documents.

Two practical constraints shaped the choices. First, a cloud model API was not available for the whole project (a free-tier quota of 20 calls per day was exhausted during Phase 2), so extraction moved to a local model. Second, the development laptop has an 8 GB GPU shared between the model and the desktop, so the model must fit with limited memory and the machine must be kept free during runs.

3.2 Design principles

Six rules were written down before the code and apply to every screen and every report:

1. **Never say "safe".** The word appears only in negated form (for example, "this does not mean it is safe").
2. **Say "not checked" when not checked.** A pair with no record, an unconfirmed medicine or a missing database each has its own status.
3. **Hold back, don't guess.** A medicine that needs confirmation is excluded from interaction checks and schedules until the user confirms it.
4. **Label what is not reviewed.** Nothing is called "reviewed" unless a real pharmacist or native speaker did the review. Machine-drafted Tamil/Hindi text is marked as a draft; the interaction supplement is marked "author checked".
5. **Keep data local.** The real flow runs on the user's computer, stores content encrypted and offers a one-click "delete my data".
6. **Report failures.** The evaluation reports raw and after-rules numbers, lists the cases that failed, and states limits next to the results.

3.3 System architecture and technology stack

CareBridge is a browser application backed by a local server. The browser handles consent, upload, review, results and schedule. The backend orchestrates extraction, normalisation, explanation and the safety check, and stores encrypted records in SQLite. The model runs in Ollama on the same machine. Figure 3.1 shows the components and Table 3.1 lists the technologies.

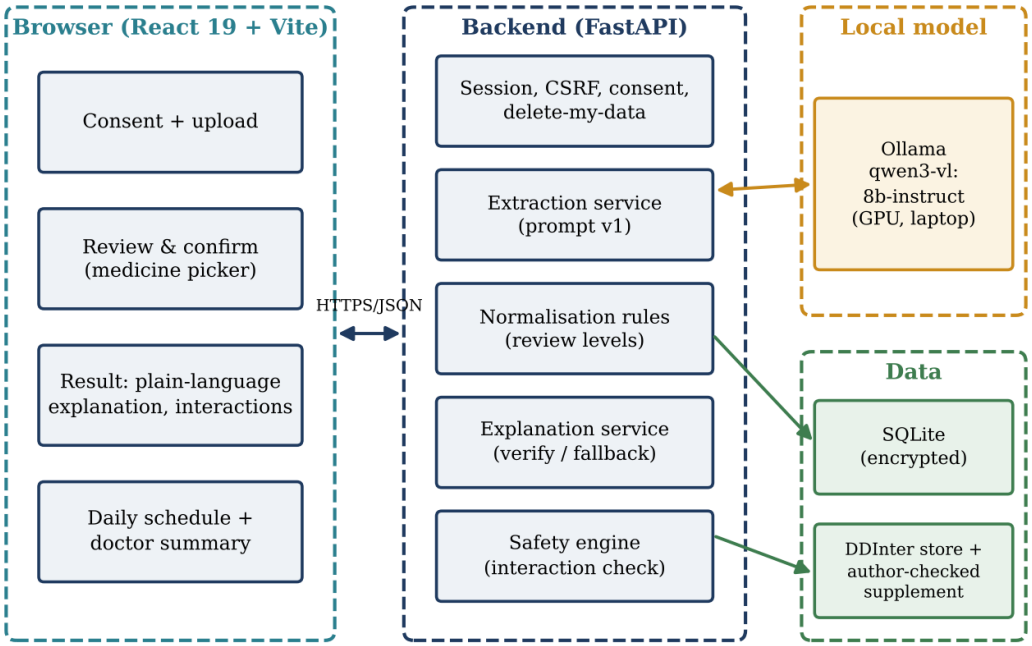


Figure 3.1: System architecture of CareBridge (all components run on the user's laptop).

Table 3.1: Technology stack

Layer	Technology	Role
Frontend	React 19, Vite, TypeScript, Tailwind CSS 4, framer-motion, GSAP	Responsive interface, animation, three languages
Backend	FastAPI (Python)	API, session, consent, orchestration
Storage	SQLite with Fernet-encrypted payloads	Per-session documents and results
Vision-language model	Qwen3-VL 8B instruct via Ollama (local)	Reads medicine rows from the photo
Interaction data	DDInter-derived store plus author-checked supplement	Pair lookup with severity and advice
Testing	pytest, vitest, Playwright, axe-core	Backend, unit, end-to-end and accessibility tests
Hosting of sample	GitHub Pages and GitHub Actions	Static, pre-recorded demonstration

3.4 Processing pipeline

Figure 3.2 shows the six steps between a photograph and a schedule. Steps 1 to 3 decide *what the document says*; steps 4 and 5 decide *what can be said about it*; step 6 turns confirmed information into a daily plan.

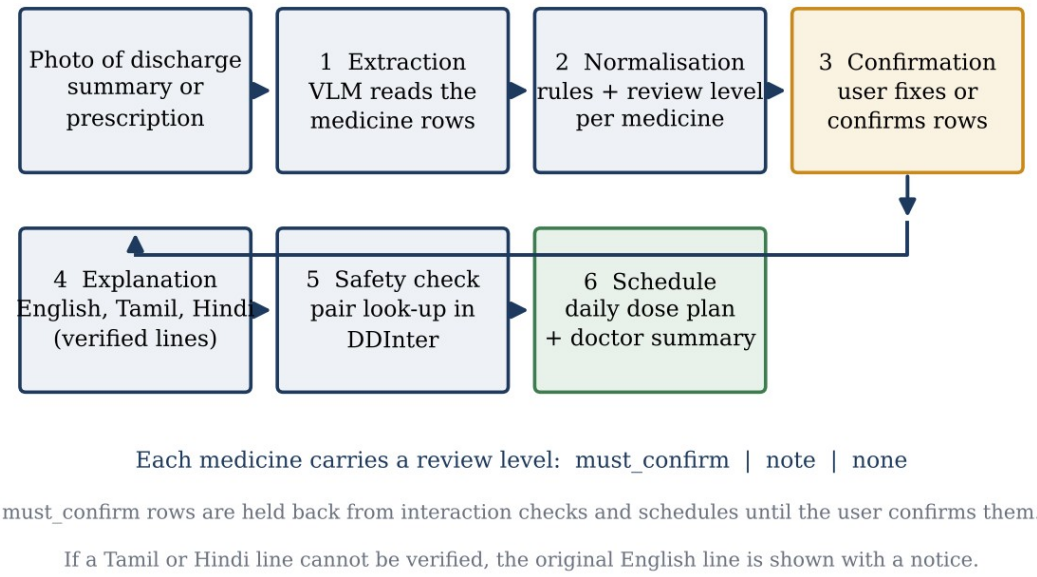


Figure 3.2: Processing pipeline from photograph to daily schedule.

- **Extraction.** The model receives the image and a fixed prompt (version 1) and returns one entry per medicine with fields such as brand name, generic name, form, strength, frequency, timing and duration, each with a short quote from the page.
- **Normalisation.** Deterministic rules fix typical model errors (for example, brand, generic and form in a single field) and assign a review level to every medicine.
- **Confirmation.** The user sees each medicine as a card. Medicines at must_confirm are resolved with a formulary picker.
- **Explanation.** Plain-language lines are produced from templates and a phrase library, with a model path that is verified line by line.
- **Safety check.** Confirmed pairs are looked up in the interaction store.
- **Schedule.** Medicines that are *schedule_ready* are placed on a daily plan, and a one-page summary for the doctor is created.

3.5 Data: synthetic documents and splits

Real discharge summaries are private, and without a ground truth they cannot be scored. The project therefore uses synthetic documents generated from templates with known medicines, strengths and instructions. Each test document is rendered under five image conditions: a clean render, plain text, a scan-like image, a phone-photo-like image and a deliberately poor-quality image. Table 3.2 lists the splits.

Table 3.2: Data splits

Split	Documents	Images	Purpose
Development (dev)	20	Dev renders	Tuning rules, prompts and templates
Held-out test	40	200 (40 × 5 conditions)	One official run under a frozen configuration
Stress set	30	60 (clean and photo renders)	Ten kinds of difficult prescription, three each

The test split comes from the same generator as the dev split. It is held out from tuning but is not independent of the generator's style, a limitation stated in the protocol. Difficulty is not balanced between splits: the test split has 16 clean, 11 abbreviated and 13 messy documents, against 5, 9 and 6 in dev.

3.6 Interaction-safety method

For every pair of medicines the safety engine returns one of four statuses (Figure 3.3). The engine first checks that both medicines are confirmed and resolved to a generic name, then that an interaction source exists for both drugs, and only then looks for a record. A pair with no record is **not_checked_no_record**; this is the commonest outcome, because 1,131 of the 2,145 pairs among the 66 formulary drugs have no DDInter or supplement record, whereas 1,014 do (998 from DDInter and 16 from the supplement). The 16-entry supplement was compiled and checked by the author against public sources; it has not been reviewed by a pharmacist.

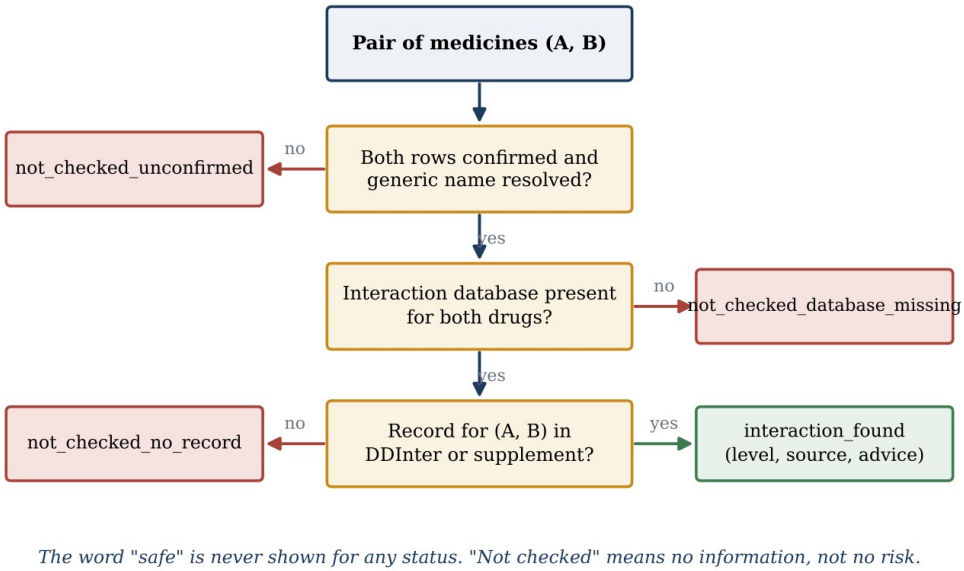


Figure 3.3: Decision flow of the interaction-safety engine. No branch ends in "safe".

3.7 Evaluation design and metrics

The evaluation was designed before the test run and written into a protocol document. The configuration was frozen with a git tag (freeze-pre-test), a manifest hashing 98 files, the model digest and a hash of the test data. The runner refuses to start in an official run if the tree is dirty, if anything has drifted, or if an official run already exists. The test split was not opened while the harness was developed: the harness tests used a synthetic fixture. Figure 3.4 summarises the design.

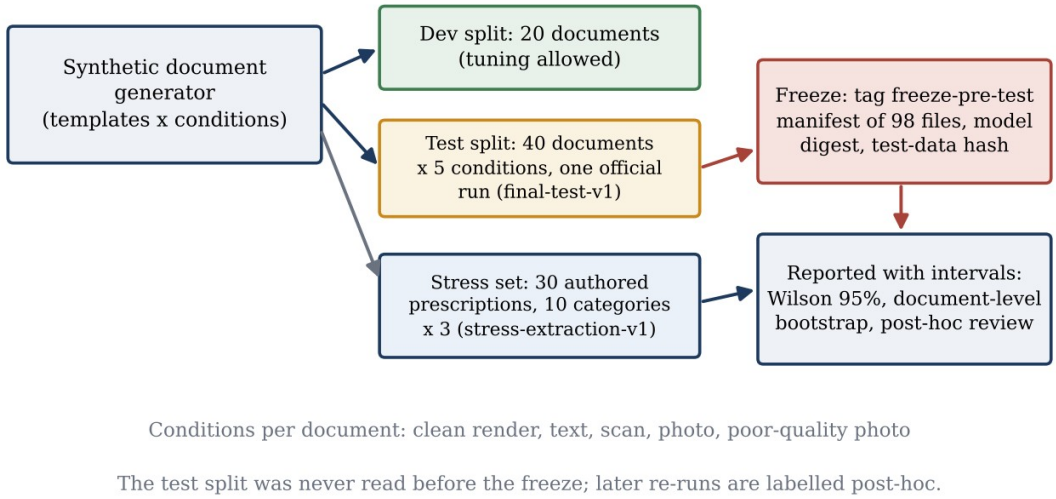


Figure 3.4: Evaluation design: three data sets, a freeze, and interval reporting.

Metrics, each reported as raw model output and after the rule layer, pooled and per condition:

- Medicine precision, recall and F1; fully-correct rate (all nine fields right); per-field accuracy; hallucinated medicines.
- Invented values by field, and name errors remaining (and how many are unflagged).
- Review burden: share of medicines at must_confirm; counts of interaction_ready and schedule_ready medicines; wrong generic names among interaction_ready medicines (target 0); wrong schedules among schedule_ready medicines.
- Safety: strict silent misses (a true interaction not shown and not covered by a flag), false alarms and the share of pairs not checked.
- Explanation: verified share per language and line kind, fall-back rate to English, and failure reasons.

Proportions use Wilson 95% intervals. F1 and the fully-correct rate also use a seeded bootstrap over documents (2,000 resamples). After the official run, a post-hoc review used a cluster-aware bootstrap and a case-by-case analysis, and is clearly labelled as post-hoc.

3.8 Privacy and security design

- Anonymous session cookie; no account or e-mail is collected.
- Request header check (CSRF) on every state-changing call.
- Consent screen with a versioned notice; consent is recorded before upload.
- Documents and results are stored with Fernet encryption in SQLite; the key is created locally and never committed.
- No disk cache of user content; “Delete my data” removes all tables for the session (measured at about 0.4 s, and shown to work even while a background explanation job is running).
- The public site contains no server and makes no /api requests.

3.9 Hosting and deployment

The real flow needs a GPU, a model and a backend, so it runs only on the author’s laptop. For public viewing, the frontend is built in a demo-only mode that replays a pre-recorded sample (a deliberately blurry scan with four medicines, including a misspelt name that exercises the picker and five interaction pairs from DDInter). GitHub Actions builds this static site, runs the unit tests, and deploys it to GitHub Pages (Figure 3.5). The privacy page of the hosted site states that GitHub may log visits under its own policy, that no cookies or analytics are used and that nothing is loaded from other sites. A separate test suite checked the live address: 78 of 78 checks passed, including zero /api requests, deep links, reload, offline behaviour, accessibility and layout in English, Tamil and Hindi.

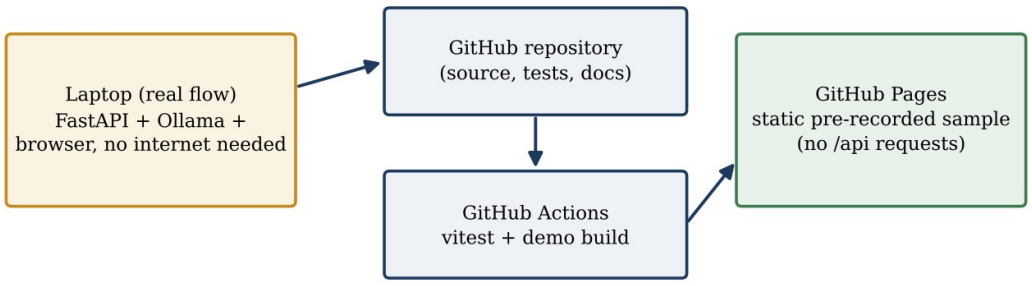


Figure 3.5: Two ways to run CareBridge: the real flow on a laptop, and the static sample on GitHub Pages.

4 FABRICATION/SIMULATION/PROGRAM OR CODE/DESIGN

This chapter describes the program: how it is organised, what each service does, and how it was tested. Short pseudocode is used where the logic matters; it is a description of the behaviour, not a copy of the source.

4.1 Repository and code organisation

The public repository (github.com/ItsTheDemiGod/carebridge, MIT licence for the project's own code, documents and synthetic data) has more than 110 commits. Its main parts are:

Folder / file	Content
backend/app	FastAPI application: routes, extraction, normalisation, explanation, safety engine, scheduler, prompts, templates, phrase library, formulary, supplement
frontend/src	React application: pages, components, i18n strings, demo mode and fixtures
data/	Synthetic documents, ground truth, templates, stress set, review sheets
eval/	Evaluation harness, frozen manifest, official results (final-test-v1), stress results, post-hoc results
scripts/	Freeze, verification, history-scan and build scripts
docs/	Protocol, results, limitations, safety, privacy, demo and decision records, screenshots
.github/workflows	Pages build and deployment workflow (no secrets)

4.2 Extraction service

The service renders the uploaded image at a maximum side of 1,280 pixels and sends it to Ollama with temperature 0 and seed 42 (context 6,144 tokens, at most 3,000 new tokens), which makes runs as repeatable as the runtime allows. The prompt asks for strict JSON with one object per medicine and, for each field, the verbatim quote from the page that supports it. The quotes are used later to detect values the model invented. The reply is parsed and validated against a schema; an unparsable reply is reported as a failure and nothing is guessed.

4.3 Normalisation rules and review levels

Raw model output is not shown to the user directly. Normalisation applies deterministic rules: it splits "brand + generic + form" fields, moves values that were placed in the wrong role, resolves names against the formulary (exact first, then close matches), and compares each value with its quote. Every medicine ends with one of three review levels and two readiness flags.

Table 4.1: Review levels, readiness and examples of flags

Item	Meaning / effect
must_confirm	The user must confirm or correct the medicine. It is excluded from interaction checks and schedules until then.

Item	Meaning / effect
note	A small inconsistency is shown as a note; the medicine is not blocked.
none	No concern raised.
interaction_ready	Name resolved to a formulary generic and confirmed; eligible for pair checks.
schedule_ready	Frequency and timing are unambiguous; eligible for a daily schedule.
unresolved_name, near_miss_name, name_not_in_quote	The name does not resolve, is close to a formulary name, or is not found in the quoted text.
duplicate_name_fields, role_corrected, incomplete_quote	A field was repaired or its evidence is incomplete; pairs are withheld until confirmed.
ambiguous_OD, weekly, unusual_strength	“OD” on an eye drop may mean once daily or right eye; weekly medicines must not get a daily schedule; unusual strengths need a human look.
conflict_timing_label	Timing label conflicts with the quote; shown as a note.

The logic of the review step can be summarised as follows (pseudocode):

```
for each medicine m in model_output:
    m = repair_fields(m)          # split brand/generic/form, fix roles
    m.generic = resolve_in_formulary(m.name)
    flags = check_against_quotes(m) + check_schedule_words(m)
    m.review = "must_confirm" if blocking(flags) else "note" if flags else
"none"
    m.interaction_ready = confirmed(m) and m.generic is not None
    m.schedule_ready    = confirmed(m) and unambiguous_frequency(m)
```

4.4 Confirmation overlay and formulary picker

For every medicine at must_confirm the interface shows the text read from the photo beside a searchable list of the 66 formulary drugs. The user picks the intended drug (or marks the row as not a medicine). The choice is stored as an overlay on the extraction, so the original model output is kept. Only after confirmation do the medicine’s pairs become checkable. In the stress test, a scripted picker choice made every garbled-name pair checkable, and only after the choice.

4.5 Explanation service with verification

Each medicine gets short lines of three kinds: advice (what it is for and how to take it), follow-up (what to ask the doctor) and warning. Lines come from templates and a phrase library; for free text a model path writes a draft in the chosen language. A verification pass compares the draft with the structured facts. A line that passes is shown; a line that fails, or whose model call is truncated, is replaced by the original English line with a notice. Tamil and Hindi strings are tagged draft_unreviewed.

4.6 Safety engine and scheduler

The safety engine implements the decision flow of Figure 3.3 over the confirmed medicines of a document. Interaction results carry the severity (major, moderate, minor or unknown), the source (DDInter or supplement) and the advice text. The scheduler places schedule_ready medicines into morning, noon, evening and night slots from the frequency and timing words, refuses to schedule weekly or as-needed medicines on a daily plan, and produces a one-page doctor summary that lists medicines, interaction findings and everything that was not checked.

4.7 User interface design

The interface was designed to look like a professional health product rather than a demo: a calm colour palette, an animated particle background, cursor-following effects and moving tiles, with real stock photography (licensed, with credits). A “calm mode” switch removes motion for users who prefer it or have vestibular sensitivities. The language switch changes all text between English, Tamil and Hindi, and the layout was checked from 320 to 1,920 pixels width (the Tamil header collapses to a menu earlier because Tamil words are longer). Contrast of all 36 text/background pairs passes the WCAG threshold used, and axe accessibility scans are clean at 1,280 and 390 pixels in all three languages.

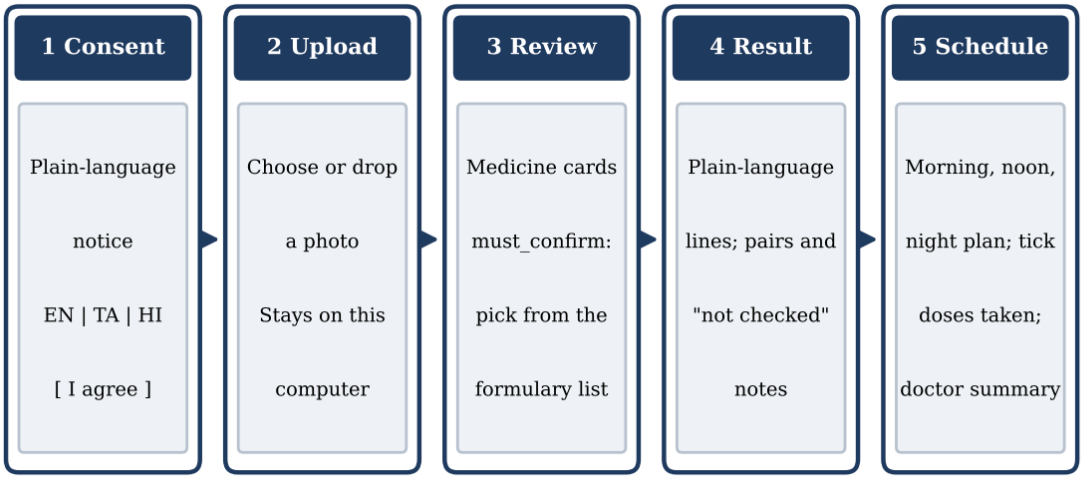


Figure 4.1: Screen flow of the real application (schematic).

4.8 Software testing and quality checks

Quality was enforced by automated tests at four levels (Table 4.2). Numbers are from the last full runs reported in the project log.

Table 4.2: Automated test suites

Suite	Result	What it covers
Backend (pytest)	More than 1,500 tests passing	Extraction parsing, rules, safety engine, scheduler, privacy, deletion, templates, wording guards
Unit (vitest)	118 passing	Front-end logic and the Pages build checks
End-to-end (Playwright, mocked)	308 passed, 14 skipped*	All screens in three languages, widths 320–1,920 px, accessibility
Hosted-build suite	75 passed locally; 78 of 78 on the live site	Zero /api requests, deep links, offline, overflow, axe
Contrast check	36 of 36 pairs pass	Colour combinations in light and calm modes
Live regression (real model)	3 flows passed (English, Tamil, Hindi with picker)	Whole chain with Ollama on the laptop

*The 14 skipped tests are screenshot-capture specs that run only on request.

Several defects were found and fixed by this process, for example a header change that hid the “Delete my data” button in Tamil on desktop widths (fixed with a visible delete icon and a test across languages and widths), a corrupted command section in the README (fixed and guarded by a test that rejects control characters), and a demo-data test whose result depended on the time of day (fixed by using a fixed, fully elapsed day).

5 RESULTS AND INTERPRETATION

5.1 Experimental setup

The official run (final-test-v1) used the frozen configuration: Qwen3-VL 8B instruct (digest 0533d743…), Ollama 0.40.5, context 6,144, at most 3,000 new tokens, image side 1,280 px, temperature 0, seed 42, extraction prompt v1 and narrative prompt v2. It ran from 02:35 to 05:00 on 20 October 2026 (about 2 h 25 min) on the author's laptop. All 200 extraction calls (40 documents × 5 conditions) were scored, nothing was skipped, and the explanation pipeline completed for all 40 documents in English, Tamil and Hindi. Two of the 320 explanation model calls were truncated (both Tamil). The SHA-256 checksums of the results folder verify, and an independent script recomputed the metrics, agreeing except for three bootstrap bounds that differ by 0.0001.

The stress extraction run (stress-extraction-v1) was a separate exercise on 30 authored prescriptions. It used Ollama 0.40.11 instead of the pinned 0.40.5; this deviation is recorded in the protocol and in the results, and the stress numbers are not claimed to be strictly comparable with the official run.

5.2 Medicine extraction

Table 5.1 gives the pooled results over the five conditions (795 medicine mentions in 200 readings). Reading the raw model output, medicine detection was already very good (F1 0.984), but only a quarter of medicine entries had every field right, because the model often filled empty fields from other text on the page or put several names in one field. After the rule layer, 88.8% were fully correct and one medicine remained that matched nothing.

Table 5.1: Pooled extraction results on the held-out test (95% intervals)

Measure	Raw model output	After rules
Medicine F1	0.984 [0.972, 0.993]	0.999 [0.996, 1.000]
Fully-correct medicine entries	0.262 [0.200, 0.328]	0.888 [0.845, 0.927]
Hallucinated medicines	13	1
Name errors remaining	–	26 of 795 (1 unflagged)
Brand-name field filled from another name on the page	276 of 313 empty fields (269 from another name)	5 of 314
Generic-name field filled when empty	62 of 211	3 of 215
Timing-label accuracy	0.734	0.990

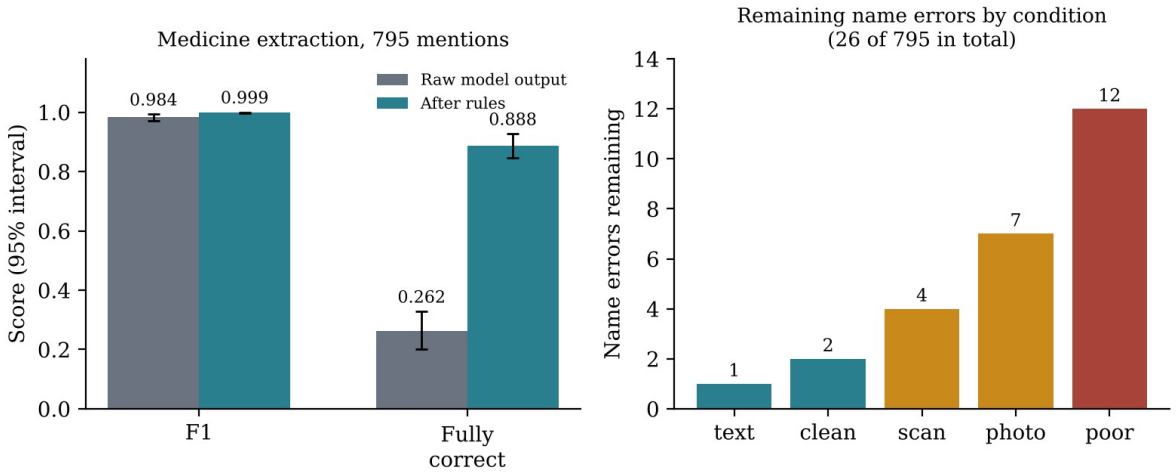


Figure 5.1: Left: raw versus after-rules extraction quality with 95% intervals. Right: name errors remaining after the rules, by image condition.

Image quality matters. After rules, the fully-correct rate ranged from 0.937 on plain text to 0.837 on the poor-quality image, and the remaining name errors rose from 1 (text) and 2 (clean) to 4 (scan), 7 (photo) and 12 (poor). Most of the 13 raw hallucinated medicines came from three documents whose layout put brand, generic and form together; the rules fixed all but one.

5.3 Review burden and readiness

The rule layer is deliberately cautious. Of the 795 medicine mentions, 51.1% were placed at must_confirm (Wilson [0.476, 0.545]; cluster bootstrap [0.436, 0.578]); 389 (48.9%) were interaction_ready and 285 (35.9%) schedule_ready. In practice, about half of the medicines on a document need one tap from the user. This is the price of not trusting the model blindly, and it is higher than a user would like; Section 6.2 lists it as a limitation.

5.4 Interaction safety

Table 5.2: Safety outcomes on the held-out test

Measure	Result
Wrong resolved generic among interaction_ready medicines	0 of 389 (rule-of-three upper bound 0.8% per reading)
False alarms (an interaction shown that is not in the ground truth)	0
Strict silent misses (true interaction not shown)	3 (all one row in one document), 0 not covered by a flag
Share of pairs “not checked”	79% to 89% depending on condition
Wrong schedules among schedule_ready medicines	2 of 285 (Wilson [0.002, 0.025]); target was 0

The most important result is the first: no wrong drug identity reached the interaction check. The three strict silent misses are not independent events; they are one misread row (“Performin” read for Metformin in a photographed document). The row was at must_confirm, so its pairs were reported as “not checked: unconfirmed” rather than hidden, which is the intended behaviour. The high “not checked” share is mostly a data-coverage limit: more than half of formulary pairs have no DDInter record, and the app says so instead of implying safety.

5.5 Explanation in three languages

Table 5.3: Explanation lines verified, by language (255 lines per language)

Language	Verified	Fell back to English	Advice	Follow-up	Warning
English	0.937	0.004	0.873	0.976	1.000
Hindi	0.667	0.275	0.754	0.762	0.516
Tamil	0.180	0.761	0.254	0.286	0.042

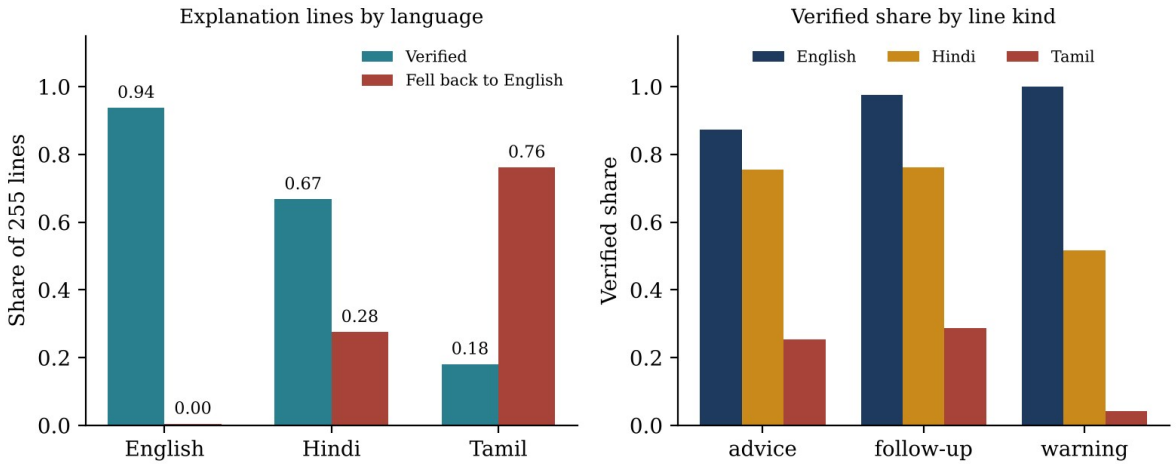


Figure 5.2: Explanation lines by language: verified versus fell back to English (left) and verified share by line kind (right).

English works well. Hindi is usable but weak on warning lines (0.516). Tamil is poor: 194 of 255 lines failed verification and were shown in English with a Tamil notice, mostly because the judge found the Tamil text “contradicts” or “omits” facts. The result is safe in the sense that no unverified Tamil is shown, but the Tamil experience is mostly English. Two caveats apply: the same model writes and judges, which can bias results [7], and no native speaker has reviewed any text. These figures should be read as a measurement of this pipeline, not of translation quality in general.

5.6 Post-hoc case review

After the official run, the failures were examined one by one (no model was re-run and the frozen results were not changed). Table 5.4 lists the cases.

Table 5.4: Failure cases found in the held-out test

Case	What happened	User-visible outcome
Wrong schedule, ds_035 (scan)	Document says SOS for an inhaler; the model read OD and the quote agreed, so no flag fired.	A daily reminder for an as-needed inhaler, no warning.
Wrong schedule, ds_038 (scan)	Quote lost “PC” from “BD PC”; timing became unspecified; only a note fired.	Dose times spread out for the antibiotic, with a note.
3 silent misses and 1 hallucination, ds_028 (photo)	“Performin” read for Metformin; row at must_confirm.	Confirm card; pairs shown as not checked.
1 unflagged name error, ds_024 (text)	Whole line “cap. racecadotril 100 mg” placed in the brand field; identity and doses correct.	Correct medicine and times; printed line in the brand slot.
2 truncated calls (Tamil)	Tamil translation chunks cut off (4 lines each in ds_024 and rx_019).	Those lines shown in English with a notice.

Two of these cases show a real design risk: when the model misreads a value and quotes its own misreading, no rule can see the difference, because the application has no independent text of the document. Likewise a note-level timing conflict does not stop a medicine from being scheduled. Both are recorded as limitations and future work, not hidden.

5.7 Development versus test comparison

Development results were better than test results on some measures and worse on others (Table 5.5). The test set was easier by mix: 45% of its medicines come from clean documents versus 26% in dev. Re-weighting to a common mix shrinks the after-rules fully-correct gap from +0.076 to about +0.044 (roughly 42% smaller) but does not remove it, and the raw brand-name gap shrinks from −0.178 to between −0.119 and −0.146. So the comparison must not be read as “no overfitting”; it shows that the headline test numbers are not inflated by tuning on test, and that the test set is somewhat easier than dev.

Table 5.5: Selected measures, dev versus test

Measure	Dev	Test	Direction on test
Brand-name accuracy (raw)	0.673	0.495	Worse
Invented-value rate (raw)	0.286	0.329	Worse
Must-confirm share	0.462	0.511	Worse (more review)
Fully correct, after rules	0.811	0.888	Better
Invented-value rate, after rules	0.056	0.022	Better
Hindi lines shown in Hindi	0.807	0.725	Worse

By difficulty (after rules, test / dev): clean 0.966 / 0.973, abbreviated 0.782 / 0.811, messy 0.870 / 0.667. The messy stratum has only 13 test and 6 dev documents, so it is noisy.

5.8 Cluster-aware intervals

Medicines in one document are not independent, so the Wilson intervals on pooled mentions are too narrow. A bootstrap that resamples whole documents (seed 20261028, 2,000 resamples) roughly doubles the width for most measures (Table 5.6). The headline conclusions do not change, but the uncertainty is larger than the official intervals suggest.

Table 5.6: Wilson versus document-level bootstrap intervals

Measure	Official Wilson	Cluster bootstrap	Width ratio
Must-confirm 0.511	[0.476, 0.545]	[0.436, 0.578]	2.04
Fully correct, raw 0.262	[0.233, 0.294]	[0.202, 0.333]	2.13
Fully correct, after rules 0.888	[0.864, 0.908]	[0.846, 0.926]	1.83
interaction_ready 0.489	[0.455, 0.524]	[0.423, 0.564]	2.02
schedule_ready 0.359	[0.326, 0.392]	[0.293, 0.432]	2.09
English lines verified	[0.901, 0.961]	[0.914, 0.961]	0.77
Tamil lines verified	[0.138, 0.232]	[0.130, 0.232]	1.09
Hindi lines verified	[0.607, 0.722]	[0.607, 0.732]	1.08

For the 2 wrong schedules in 2 documents, the bootstrap is unreliable (too few events), so the Wilson upper bound of 0.025 is used. For the wrong-generic result (0 of 389 readings, which are 106 distinct medicines in 39 documents), the exact one-sided 95% upper bound across documents is 7.4%: with 40 documents the test cannot rule out a few percent of documents with such an error.

5.9 Stress test on authored prescriptions

The stress set has ten kinds of difficult prescription, three of each: a major-interaction pair, a moderate pair, a supplement pair, a duplicate, a garbled name, a weekly medicine, an unusual strength, an eye drop with OD, an eight-medicine list and a control. Among the 66 formulary drugs, the 2,145 pairs contain 17 major, 261 moderate, 44 minor and 692 pairs with a record of unknown severity (1,131 pairs have no record at all). The four pass criteria were fixed before the run, and all were met (Figure 5.3, right).

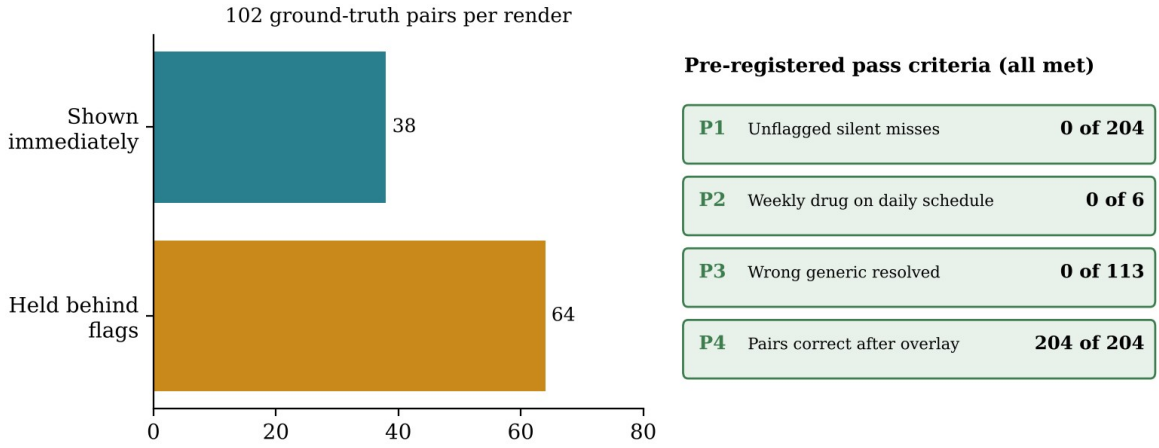


Figure 5.3: Stress test. Left: of 102 ground-truth interaction pairs per render, 38 were shown immediately and 64 were held behind confirmation flags. Right: the four pre-registered criteria.

Table 5.7: Stress extraction behaviour by category (both renders)

Category	Result
Medicine detection	104 of 104 medicines in each render; 0 hallucinated
Duplicates	All 3 flagged in each render
Weekly medicines	All 3 flagged, none scheduled daily
Eye drops with OD	ambiguous_OD fired on all 3; none scheduled
Unusual strengths	Flagged on all 3
Garbled names	3 of 3 flagged on the clean render; 2 of 3 on the photo render
False alarms	0 before confirmation; 2 after the scripted picker on one photo (st_15)

The weakest result is that only 38 of 102 pairs (37.3%, Wilson [0.285, 0.469]; document bootstrap [0.132, 0.631]) were shown immediately. The other 64 were held behind flags that fired on *correctly read* medicines (duplicate_name_fields, role_corrected, incomplete_quote). None was lost silently; they become checkable after confirmation. With perfect extraction, the engine would show 100 of 102 pairs, so the loss is due to extraction caution, not engine blindness. The one unflagged garbled name was “Ibuirofen”, which the model silently corrected to “Ibuprofen” even in its quote; an unrelated flag held the row back, but this kind of quote-consistent correction is invisible to the rules. Seventy-eight of the 102 pairs come from only three long-list documents, so all these intervals are wide.

5.10 Interpretation

- **Layering works.** The model alone had F1 0.984 but only 26% fully correct entries; the rules plus confirmation raised this to 89% and kept wrong drug identities out of the safety check.

- **Caution has a cost.** About half of the medicines need user confirmation, and over three quarters of pairs end as “not checked”. The app is honest, but not yet convenient.
- **Language support is uneven.** English is reliable, Hindi is partial and Tamil mostly falls back to English. This is a finding about the pipeline and the model, and a clear target for native-speaker review.
- **Some errors cannot be caught by rules.** A misread that agrees with its own quote passes every check. Only an independent source, such as a second reading or a barcode, would catch it.
- **Evidence is limited.** All tests used synthetic documents of one generator, and 40 documents give wide intervals, so the results show the prototype behaves as designed, not that it is clinically safe.

6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

CareBridge shows that a post-discharge medicine companion can be built with an open vision-language model on a single laptop, in three languages, and with safety behaviour designed around uncertainty. A photograph is turned into a medicine list, a plain-language explanation, an interaction check and a daily schedule, and at each step the system is explicit about what it did not or could not check. On 40 held-out synthetic documents it reached medicine F1 of 0.999 after the rule layer, 88.8% fully correct entries, no wrong generic name among 389 interaction-ready readings, and no unflagged silent miss; all four pre-registered stress criteria were met.

Equally important are the results that fell short: two wrong schedules, about half of the medicines needing confirmation, over three quarters of pairs “not checked”, and Tamil explanations that mostly fall back to English. Reporting these openly, with raw and after-rules numbers, a frozen protocol and a post-hoc case review, is part of the contribution. The objectives listed in Section 1.3 have all been met at prototype level, and the work is published as a verified, pre-recorded sample on GitHub Pages together with the full source under the MIT licence.

6.2 Limitations

- Synthetic data from a single generator; no real patient documents; real-photo testing and usability sessions were not carried out.
- Not clinically validated; not a medical device. The author-checked interaction supplement and all Tamil/Hindi text lack pharmacist and native-speaker review.
- The same model writes and judges explanations; a native speaker would be needed to measure true quality.
- Misreadings that agree with their own quote, and note-level timing conflicts, can reach the schedule without a block.
- Output of the local model is not reproducible across Ollama versions (0.40.5 versus 0.40.11 gave different flags on a check of five documents).
- About half of the medicines need confirmation, and 53% of formulary pairs have no record in DDInter or the supplement.
- Test and dev share one generator and differ in difficulty mix; with 40 documents the intervals are wide.

6.3 Future scope

1. Run the planned real-photo test (printed documents photographed under five conditions) and usability sessions with at least five participants, using the prepared protocol, tasks, consent forms and SUS questionnaire.

2. Obtain pharmacist review of the interaction supplement and native-speaker review of the Tamil and Hindi strings (review sheets are already prepared), and then measure Tamil verification again.
3. Add an independent check against misreadings: a second reading pass, comparison with the form against the frequency (for example an inhaler with “OD”), or barcode scanning.
4. Make a timing conflict block scheduling until confirmed, and reduce unnecessary confirmations by refining the flags that fire on correctly read medicines.
5. Extend interaction coverage with additional open sources and show clearly which source supports each statement.
6. Add handwriting-heavy test documents, and evaluate smaller or quantised models for devices without a GPU.
7. Stratify the data splits by difficulty and template, and add an independent test set not produced by the same generator.
8. Explore a pilot with a hospital pharmacy, under proper ethics approval and data-protection review, before any real patient use.

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APPENDIX: Abbreviations and project links

Table A.1: Abbreviations

Term	Meaning
OD	Once daily (on eye drops it can also mean the right eye, which is why the app flags it)
BD	Twice daily
PC	After food
SOS	Take only if needed
VLM	Vision-language model: a model that reads an image and a text prompt
DDI	Drug–drug interaction
DDInter	Open drug–drug interaction database used as the main source [4]
Ollama	Software that runs language models on the local computer [11]
F1	Combined score of precision and recall for finding the right medicines
Wilson interval	Confidence interval for a proportion [8]
Cluster bootstrap	Resampling of whole documents to estimate uncertainty [9]
must_confirm / note / none	Review levels given to each medicine
interaction_ready / schedule_ready	Medicine is eligible for interaction checks / a daily schedule
SUS	System Usability Scale questionnaire (prepared, not yet run)

Table A.2: Project links

Item	Location
Source code (MIT)	https://github.com/ItsTheDemiGod/carebridge
Pre-recorded sample site	https://itsthedemigod.github.io/carebridge/
Official test results	eval/results/final-test-v1 (with SHA256SUMS)
Protocol and results documents	docs/FINAL_TEST_PROTOCOL.md, FINAL_TEST_RESULTS.md, FINAL_TEST_POSTHOC.md, STRESS_RESULTS.md, LIMITATIONS.md
Third-party licences	THIRD_PARTY_NOTICES.md (DDInter excerpt CC BY-NC-SA; photos and fonts keep their own licences)

Disclaimer. CareBridge is a class-project prototype. It is not a medical device, has not been clinically validated and must not be used to make treatment decisions. Always follow the instructions of your doctor or pharmacist.